Executive Summary: Regression Analysis

TikTok claims classification project

OVERVIEW

TThis project evaluates a logistic regression model designed to predict account verification status (`verified` vs. `not verified`) using video engagement metrics and behavioral features. The primary goal is to help streamline content moderation workflows by automatically categorizing and prioritizing user reports to manage the review backlog more efficiently.

PROJECT STATUS

Cleaned the dataset by dropping 298 missing values and examining key feature distributions (e.g., video_duration_sec) to ensure model readiness.

Model Baseline Established: Trained and evaluated a initial logistic regression model on a balanced test set of 7,154 records.

Initial Results: Achieved an overall classification accuracy of **63%**. The model currently demonstrates strong precision for verified accounts (74%) and high recall for non-verified accounts (87%)

NEXT STEPS

Model Iteration & Feature Engineering: Explore non-linear algorithms (e.g., Random Forest or XGBoost) and create composite engagement ratios to improve overall F1-scores and reduce false positive rates.

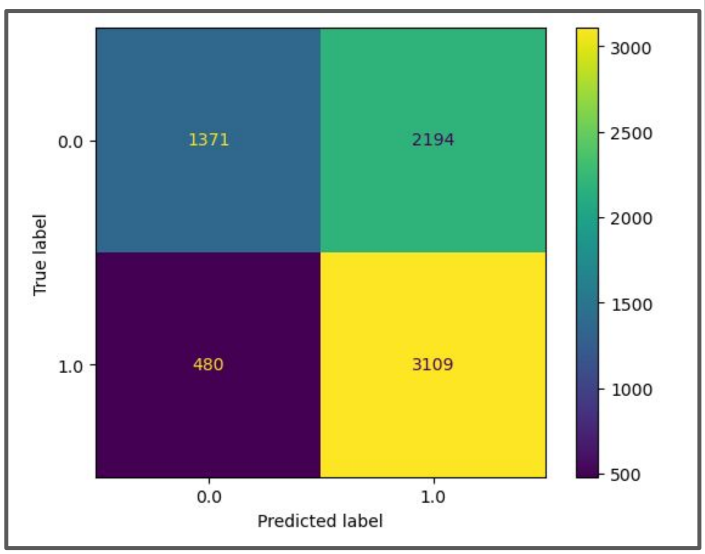
Threshold Tuning: Adjust decision thresholds to better balance recall and precision for the "verified" class based on moderation priorities.

KEY INSIGHTS

BBehavioral Distinction: Accounts posting opinion content are significantly more likely to be verified compared to those publishing unverified claims.

Model Bias Towards "Not Verified": While the model catches 87% of unverified accounts, it misses a substantial portion of verified accounts (38% recall), indicating room to refine classification thresholds.

Confusion matrix for logistic regression model



Upper-left: the number of videos posted by unverified accounts. Upper-right: the number of videos posted by unverified accounts. Lower-left: the number of videos posted by verified accounts. Lower-right: the number of videos posted by verified accounts.